

The tendency of a chemical species to undergo reduction during an electrochemical reaction is indicated by the standard potential E° for the reaction. As represented generally by Reaction 1, E° measures the potential Z at which element X with oxidation number N accepts N electrons and is reduced to its elemental state.



Reaction 1

In a galvanic cell, $E^\circ > 0$ indicates a spontaneous reaction. Therefore, chemical species with higher positive E° values are reduced more easily than those with lower E° values.

Using E° as a basis of comparison, the relative thermodynamic stabilities of different chemical species involving the same element at different oxidation states can be presented graphically using a Frost diagram. For a series of compounds containing element X , a Frost diagram plots the value of NE° for each compound against the corresponding oxidation number N of element X within the species. Frost diagrams for several species containing manganese or chlorine are shown in Figure 1.

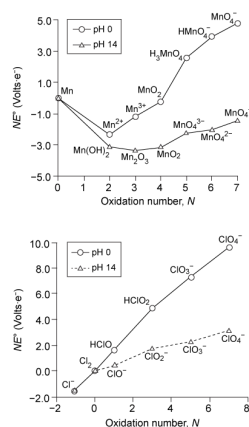


Figure 1 Frost diagrams for manganese and chlorine

On a Frost diagram, NE° is proportional to the standard Gibbs free energy ΔG° according to the relationship

$$\Delta G^\circ = -FNE^\circ = -nFE^\circ$$

Equation 1

where n is the number of moles of electrons transferred during the electrochemical process and F is the Faraday constant. Free energy considerations also show that a species is prone to disproportionation if its position on the Frost diagram lies above a line connecting the points of two adjacent species, as seen for species B in Figure 2.

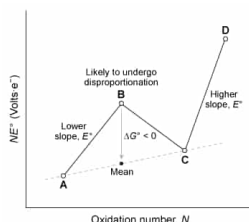


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

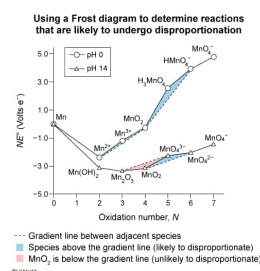
The slope of a line segment joining two species on a Frost diagram is equal to the standard reduction potential for the couple. A greater slope indicates a higher corresponding reduction potential. As a result, in Figure 2, the reduction potential for B to A is lower than that for D to C.

Based on the passage, which of the following species is LEAST likely to undergo a disproportionation reaction?

- ☐ A. H_3MnO_4 , pH = 0 [13%]
- ☐ B. MnO_4^{3-} , pH = 14 [12%]
- ☐ C. Mn^{3+} , pH = 0 [12%]
- ☒ D. MnO_2 , pH = 14 [61%]

Correct
60% answered correctly

Explanation:



A **disproportionation reaction** is a redox reaction in which both the **oxidation and reduction** occur to atoms of the **same element**. Disproportionation reactions can be easily identified by **assigning oxidation numbers** and then **comparing** the oxidation number of the given element in the reactants with the oxidation number of the same element in the products. If a compound undergoes disproportionation, some of the compound molecules supply atoms of a particular element for oxidation, and other molecules of the compound supply atoms of the **same element** for reduction.

Exam questions sometimes test required topics by presenting the concepts in an unfamiliar context using details that are not required knowledge but are explained within the passage. In this passage, Frost diagrams are presented as a way of predicting which chemical species are likely to undergo disproportionation. As shown in Figure 2 of the passage, a species is likely to undergo disproportionation if its position on the Frost diagram lies **above a line** connecting the points of **two adjacent species** (ie, the points on each side of the point of interest) because $\Delta G^\circ < 0$ with respect to the mean of the potential gradient.

Accordingly, answering this question is a matter of graphical evaluation. Based on the above criteria and an examination of Figure 1, MnO_2 is least likely to undergo disproportionation because its point on the Frost diagram lies **below** the line connecting the points of two neighboring species.

(Choices A and B) On the Frost diagram in the plot for pH = 0, H_3MnO_4 is positioned above a dashed line connecting MnO_2 and HMnO_4^- . Likewise, MnO_4^{3-} is above a dashed line connecting MnO_2 and MnO_4^{2-} . Therefore, in both cases, $\Delta G^\circ < 0$ relative to the mean, and these species are *likely* to undergo a disproportionation reaction.

(Choice C) The points for Mn^{2+} , Mn^{3+} , and MnO_2 are not perfectly linear, and Mn^{3+} lies slightly above the dashed line connecting Mn^{2+} and MnO_2 . Although the tendency toward disproportionation is low due to the small difference, Mn^{3+} is not the least likely to undergo disproportionation.

Educational objective:

A disproportionation reaction is a redox reaction in which both the oxidation and the reduction occur with atoms of the same element. The tendency of a reaction toward disproportionation can be evaluated graphically.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402386
System:
Atomic Theory and Chemical Composition

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Reaction 1

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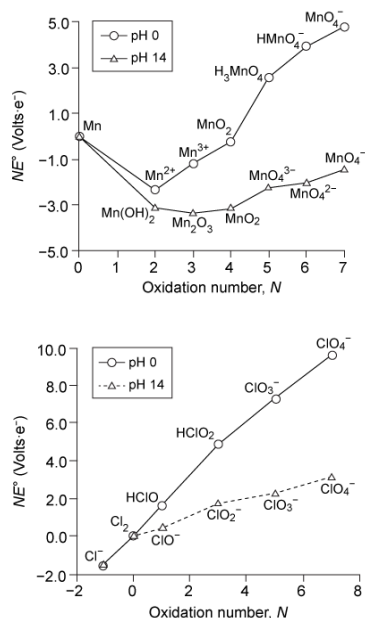


Figure 1 Frost diagrams for manganese and chlorine

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Equation 1

where n is the number of moles of electrons transferred during the electrochemical process and F is the Faraday constant. Free energy considerations also show that a species is prone to disproportionation if its position on the Frost diagram lies above a line connecting the points of two adjacent species, as seen for species B in Figure 2.

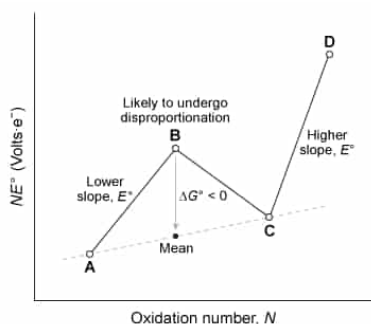
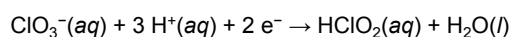


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

The slope of a line segment joining two species on a Frost diagram is equal to the standard reduction potential for the couple. A greater slope indicates a higher corresponding reduction potential. As a result, in Figure 2, the reduction potential for B to A is lower than that for D to C.

Based on the slope of the corresponding line segment in Figure 1, what is the standard potential E° for the half-reaction below?



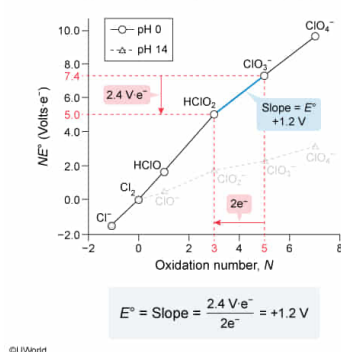
- ☐ A. +2.4 V [33%]
☒ B. +1.2 V [45%]
☐ C. +0.3 V [9%]
☐ D. -0.6 V [11%]

Correct

45% answered correctly

Explanation:

The slope of the line segment between two species on a Frost diagram equals E° for the couple



As stated in the passage, the **slope of a line segment** joining two species on a Frost diagram equals the **standard reduction potential** E° for the couple. Accordingly, the slope between two points on an x,y-coordinate system is found by dividing the change in the y-axis values by the change in the x-axis values:

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{(y_{\text{final}} - y_{\text{initial}})}{(x_{\text{final}} - x_{\text{initial}})}$$

In the given half-reaction, $\text{ClO}_3^-(aq)$ is **reduced** by accepting two electrons to form $\text{HClO}_2(aq)$ under *acidic conditions* (as indicated by the $3 \text{H}^+(aq)$ in the reactants). The point on the Frost diagram (Figure 1) corresponding to ClO_3^- under acidic conditions (pH 0) therefore represents the initial condition of the half-reaction, whereas the point corresponding to HClO_2 represents the final condition.

The change in oxidation number (Δx) between ClO_3^- and HClO_2 equals the number of electrons transferred for the change in NE° (Δy), and the **slope** of the line segment between these points equals the standard potential for the reduction. Substituting the x- and y-coordinate values for the ClO_3^- and HClO_2 points on the Frost diagram into the slope equation above gives:

$$\text{Slope} = \frac{\Delta NE^\circ}{\Delta N} = \frac{(5.0 - 7.4) \text{ V} \cdot e^-}{(3 - 5)e^-} = \frac{-2.4 \text{ V} \cdot e^-}{-2 e^-} = +1.2 \text{ V}$$

Therefore, the Frost diagram for chlorine indicates that the standard potential E° for the reduction of ClO_3^- to HClO_2 is +1.2 V.

(Choice A) A value of +2.4 V · e⁻ is obtained if the change in NE° (y-axis) is not divided by the change in the oxidation state N (x-axis). The slope ($\Delta y/\Delta x$), not Δy , is equal to E° for the reduction, as confirmed by the units.

(Choice C) A standard potential of +0.3 V is required for the reduction of ClO_3^- to ClO_2^- , which occurs only under *basic conditions* (pH 14). The given half-reaction occurs under acidic conditions (ie, there is H^+ in the reactants) and forms an acidic compound (HClO_2).

(Choice D) A value of -0.6 V · e⁻ results if the change in NE° (y-axis) calculated under basic conditions for the reduction of ClO_3^- to ClO_2^- is not divided by the change in N (x-axis). A calculation error gives a negative slope.

Educational objective:

The standard reduction potential E° for an oxidation-reduction couple can be evaluated graphically using electrochemical data. The

slope of a line segment between two points on an x,y -coordinate system is found by dividing the change in the y -axis values by the change in the x -axis values.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402387

System:
Atomic Theory and Chemical Composition

The tendency of a chemical species to undergo reduction during an electrochemical reaction is indicated by the standard potential E° for the reaction. As represented generally by Reaction 1, E° measures the potential Z at which element X with oxidation number N accepts N electrons and is reduced to its elemental state.



Reaction 1

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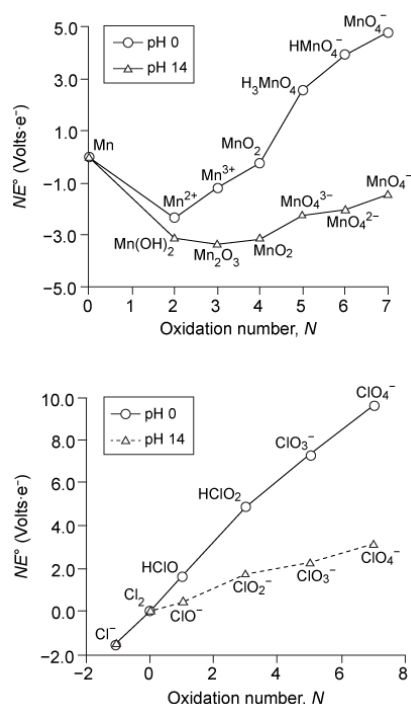


Figure 1 Frost diagrams for manganese and chlorine

On a Frost diagram, NE° is proportional to the standard Gibbs free energy ΔG° according to the relationship

$$\Delta G^\circ = -FNE^\circ = -nFE^\circ$$

Equation 1

where n is the number of moles of electrons transferred during the electrochemical process and F is the Faraday constant. Free energy considerations also show that a species is prone to disproportionation if its position on the Frost diagram lies above a line connecting the points of two adjacent species, as seen for species B in Figure 2.

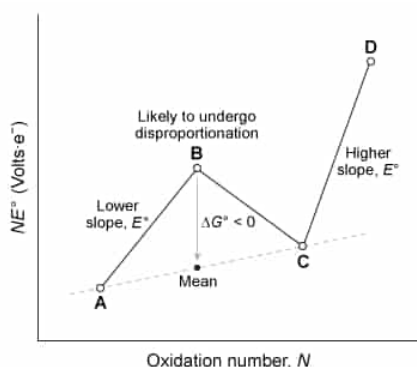
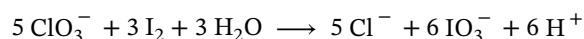


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

The slope of a line segment joining two species on a Frost diagram is equal to the standard reduction potential for the couple. A greater slope indicates a higher corresponding reduction potential. As a result, in Figure 2, the reduction potential for B to A is lower than that for D to C.

Iodine reacts with ClO_3^- under acidic conditions according to the reaction shown below.



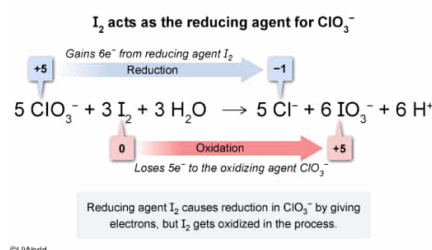
Does I_2 act as the oxidizing agent or reducing agent in the reaction?

- ☐ A. Oxidizing agent, because I_2 is reduced [15%]
☒ B. Reducing agent, because I_2 is oxidized [72%]
☐ C. Oxidizing agent, because ClO_3^- is reduced [7%]
☐ D. Reducing agent, because ClO_3^- is oxidized [4%]

Correct

72% answered correctly

Explanation:



In an electrochemical reaction, **oxidation and reduction** always occur simultaneously. The electrons lost by the atom that is oxidized are gained by another atom that is reduced. As a result, the chemical species containing the atom being oxidized is the **reducing agent** that *causes reduction* in another atom by giving up the electrons involved in the reduction in the other atom. Conversely, the species with an atom that is reduced functions as the **oxidizing agent** that *causes oxidation* by taking electrons from (oxidizing) another atom.

In the given reaction, the chlorine atom in ClO_3^- is reduced while the iodine atoms in I_2 are oxidized. This can be determined by examining the changes in the **bonds to oxygen** between the species, by assigning **oxidation numbers**, or by reading the oxidation numbers for chlorine directly from the Frost diagram in the passage (Figure 1). During the reaction, the chlorine atoms lose bonds to oxygen and have a decrease in oxidation state from +5 to -1. In contrast, the iodine atoms gain bonds to oxygen and have an increase in oxidation state from 0 to +5.

Therefore, I_2 acts as the reducing agent by giving up electrons and causing ClO_3^- to be reduced. In the process, I_2 itself gets oxidized because it is losing the electrons.

(Choices A and C) Oxidizing agents *cause oxidation* in another compound. I_2 cannot be the oxidizing agent in the given reaction because the chlorine atom in ClO_3^- is *reduced*. It is the I_2 itself that is oxidized.

(Choice D) Reducing agents *cause reduction* in another compound. Because I_2 is the reducing agent in the given reaction, ClO_3^- is *reduced* rather than oxidized.

Educational objective:

A reducing agent is a chemical species that gets oxidized and causes reduction in another atom in the process, whereas an oxidizing agent is a chemical species that gets reduced and causes oxidation in another atom as a result. Oxidation and reduction always occur together.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402388
System:
Atomic Theory and Chemical Composition

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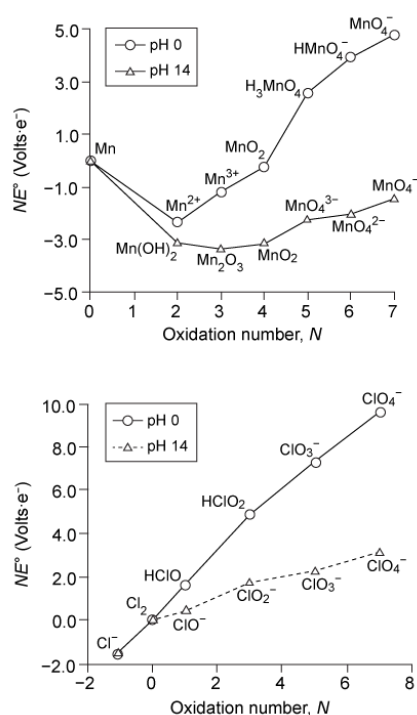


Figure 1 Frost diagrams for manganese and chlorine

On a Frost diagram, NE° is proportional to the standard Gibbs free energy ΔG° according to the relationship

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Equation 1

where n is the number of moles of electrons transferred during the electrochemical process and F is the Faraday constant. Free energy considerations also show that a species is prone to disproportionation if its position on the Frost diagram lies above a line connecting the points of two adjacent species, as seen for species B in Figure 2.

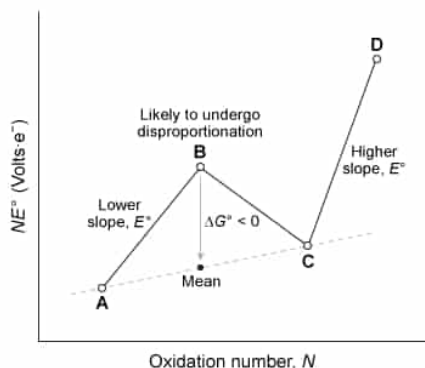


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

The slope of a line segment joining two species on a Frost diagram is equal to the standard reduction potential for the couple. A greater slope indicates a higher corresponding reduction potential. As a result, in Figure 2, the reduction potential for B to A is lower than that for D to C.

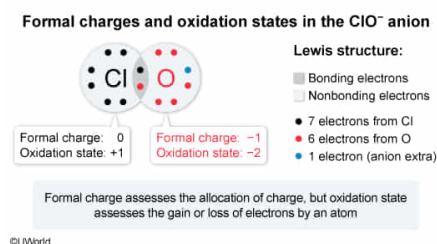
Formal charge and oxidation state are two different ways of accounting for electrons between atoms. Based on the Lewis structure, which of the following pairs of values gives the formal charge and oxidation state of the chlorine atom in the ClO^- anion?

- ☐ A. Formal charge = 0; Oxidation state = 0 [4%]
- ✓ ☒ B. Formal charge = 0; Oxidation state = +1 [64%]
- ☐ C. Formal charge = 0; Oxidation state = -1 [19%]
- ☐ D. Formal charge = -1; Oxidation state = -2 [11%]

Correct

64% answered correctly

Explanation:



The **oxidation state** (oxidation number) of an atom is conceptually **different from the formal charge** of that atom, and the two values are **often not the same number**. **Formal charge** assesses the **allocation of charge** resulting from an atom's bonding configuration by assuming that all bonding electrons are equally shared (divided) between each pair of atoms.

$$\text{Formal charge} = \left(\text{Group valence} \right) - \left(\text{Nonbonding electrons} \right) - \frac{1}{2} \left(\text{Bonding electrons} \right)$$

In contrast, the **oxidation state** tracks the **gain or loss of electrons** relative to an atom's **valence number** by **assigning all bonding electrons** to the most **electronegative** atom of the bond. When the atoms in the bond are the same, each atom is assigned 1 electron per bond.

$$\text{Oxidation state} = \left(\text{Group valence} \right) - \left(\text{Nonbonding electrons} \right) - \left(\text{Assigned bonding electrons} \right)$$

Applying these formulas to the Lewis structure of ClO^- , the formal charge and oxidation state of the chlorine atom are calculated as follows:

$$\text{Formal charge (Cl)} = (7)_{\text{Group valence}} - (6)_{\text{Nonbonding electrons}} - \frac{1}{2}(2)_{\text{Bonding electrons}} = 0$$

$$\text{Oxidation state (Cl)} = (7)_{\text{Group valence}} - (6)_{\text{Nonbonding electrons}} - (0)_{\text{Assigned bonding electrons}} = +1$$

(Choice A) Although the Cl atom has zero formal charge, the Cl atom has an oxidation state of +1 because it gives up 1 electron to the more electronegative O atom during bond formation.

(Choice C) An incorrect value of -1 for the oxidation state of Cl is obtained by assuming that Cl is more electronegative than O and wrongly assigning 2 bonding electrons to Cl during the calculation. The O atom does not lose electrons to Cl.

(Choice D) The O atom in the ClO^- anion has a formal charge of -1 and an oxidation state of -2.

$$\text{Formal charge (O)} = (6)_{\text{Group valence}} - (6)_{\text{Nonbonding electrons}} - \frac{1}{2}(2)_{\text{Bonding electrons}} = -1$$

$$\text{Oxidation state (O)} = (6)_{\text{Group valence}} - (6)_{\text{Nonbonding electrons}} - (2)_{\text{Assigned bonding electrons}} = -2$$

Educational objective:

Formal charge assesses the allocation of charge to an atom by dividing bonding electrons evenly between bonded atoms. Oxidation state assesses the gain or loss of electrons relative to an atom's valence number by assigning all bonding electrons to the more electronegative atom.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402389

System:
Atomic Theory and Chemical Composition

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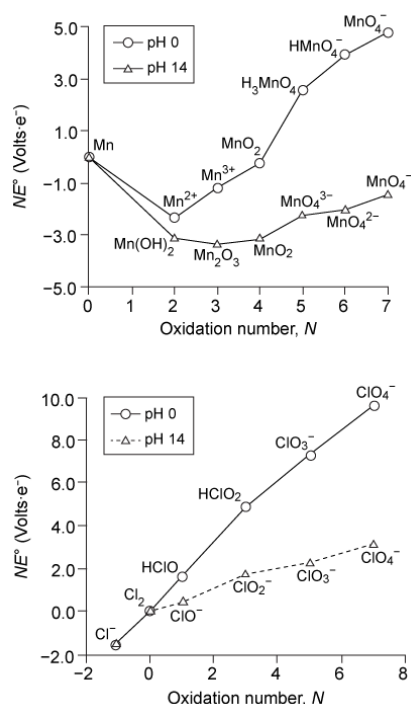


Figure 1 Frost diagrams for manganese and chlorine

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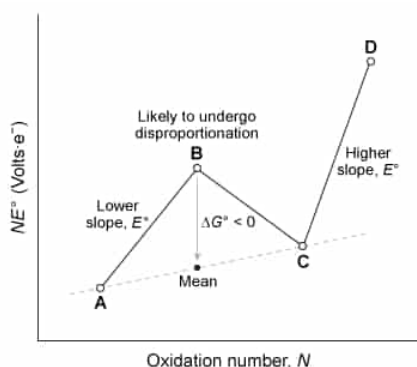
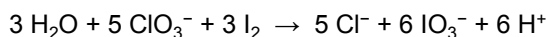


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

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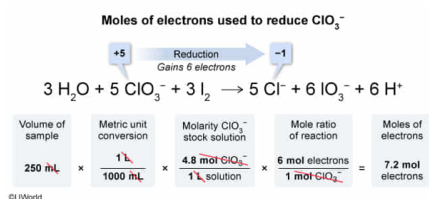
Consider the reaction of iodine with ClO_3^- under acidic conditions, as shown above. If 6 moles of electrons are required to reduce 1 mole of ClO_3^- , how many moles of electrons are required to completely reduce all the ClO_3^- ions present in 250 mL of a 4.8 M ClO_3^- solution?

- ✓ ☒ A. 7.2 moles [74%]
☐ B. 6.0 moles [8%]
☐ C. 4.8 moles [8%]
☐ D. 1.2 moles [8%]

Correct

74% answered correctly

Explanation:



The number of **moles** in the sample taken from a stock solution can be calculated by multiplying the **molarity (mol/L)** of the stock solution by the **volume (L)** of the sample. The moles of the reactants and products of a reaction are related to one another by the **mole ratios** from the balanced reaction equation. Mole ratios can also be applied to electrochemical processes because electrons are gained or lost in definite proportions.

In this question, 250 mL (0.25 L) of a 4.8 M ClO_3^- stock solution supplies 1.2 mol of ClO_3^- for the reaction.

$$250 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{4.8 \text{ mol ClO}_3^-}{1 \text{ L solution}} = 1.2 \text{ mol ClO}_3^-$$

In the given reaction, ClO_3^- is converted into Cl^- . As seen from Figure 1 in the passage, this involves the reduction of the chlorine atom from an oxidation number of +5 in ClO_3^- to an oxidation number of -1 in Cl^- (a difference of 6 units). Because a change of 1 unit in the oxidation number corresponds to a transfer of 1 electron, 6 electrons are needed for each ClO_3^- anion that is reduced. Using this ratio, the moles of electrons used in the reaction of 1.2 mol ClO_3^- can be calculated.

$$1.2 \text{ mol ClO}_3^- \times \frac{6 \text{ mol electrons}}{1 \text{ mol ClO}_3^-} = 7.2 \text{ mol electrons}$$

(Choice B) In the reaction, 6.0 moles of electrons are required for 1 mole of ClO_3^- , but there are 1.2 moles of ClO_3^- in 250 mL of a 4.8 M ClO_3^- solution.

(Choices C and D) A quantity of 4.8 moles of ClO_3^- is present in 1 L of a 4.8 M ClO_3^- solution, but only 1.2 moles of ClO_3^- are present in the 0.25 L sample used for the reaction. Nevertheless, the question asks for the moles of electrons rather than the moles of ClO_3^- .

Educational objective:

The number of moles transferred in a sample taken from a stock solution can be calculated by multiplying the molarity of the stock solution by the sample volume. Molar ratios taken from balanced chemical reactions or electrochemical

processes can be used to relate the moles of one species to the moles of another.

Time spent:0 sec

Subject:
General Chemistry

QID:402390

System:
Atomic Theory and Chemical Composition

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Reaction 1

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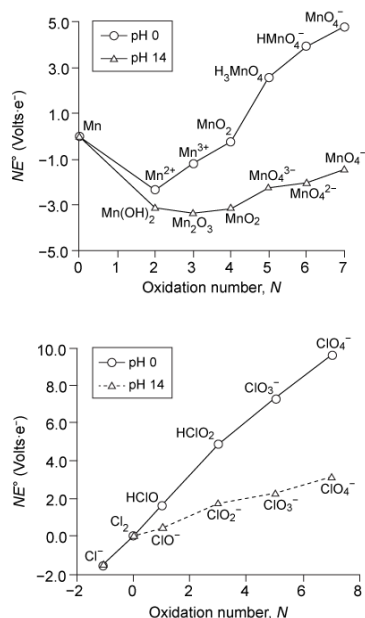


Figure 1 Frost diagrams for manganese and chlorine

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Equation 1

where n is the number of moles of electrons transferred during the electrochemical process and F is the Faraday constant. Free energy considerations also show that a species is prone to disproportionation if its position on the Frost diagram lies above a line connecting the points of two adjacent species, as seen for species B in Figure 2.

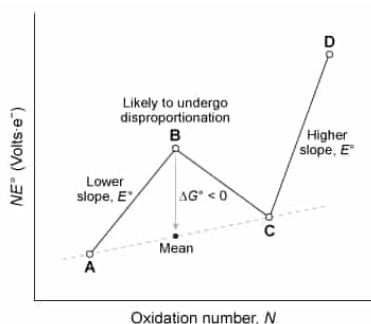


Figure 2 General Frost diagram for an element forming species A, B, C, and D.

The slope of a line segment joining two species on a Frost diagram is equal to the standard reduction potential for the couple. A greater slope indicates a higher corresponding reduction potential. As a result, in Figure 2, the reduction potential for B to A is lower than that for D to C.

Suppose an experiment is to be performed utilizing a reaction that reduces ClO_4^- to ClO_3^- . Which experimental approach can be expected to make the reduction of ClO_4^- more favorable during the reaction?

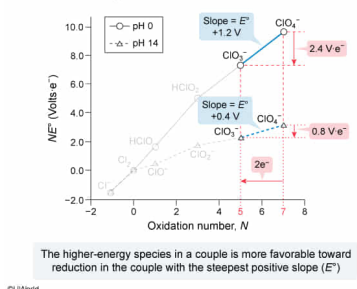
- ✔ ☒ A. Perform the reaction under acidic conditions [62%]
☐ B. Perform the reaction under neutral conditions [3%]
☐ C. Perform the reaction under basic conditions [29%]
☐ D. The pH of the reaction mixture does not make a difference [4%]

Correct

62% answered correctly

Explanation:

Reduction of ClO_4^- is more favorable under acidic conditions



As stated in the passage, an NE° (the y-axis) value on a Frost diagram is proportional to the change in the standard Gibbs free energy ΔG° according to the relationship

$$\Delta G^\circ = -FNE^\circ = -nFE^\circ$$

where n is the number of moles of electrons transferred during the electrochemical process, F is the Faraday constant, and E° is the **standard reduction potential**. Because reactions with a $\Delta G^\circ < 0$ can occur spontaneously, a redox process with a larger positive E° is more favorable than a redox process with a smaller E° . The passage states that on a Frost diagram, the slope of a line segment joining two species is equal to E° for the couple. As a result, when comparing line segments, the segment with the **greatest slope** is the **more favorable** reaction.

Comparing the slopes of the line segments between ClO_4^- and ClO_3^- for both pH 0 and pH 14 on the Frost diagram of chlorine (Figure 1 in the passage), the slope for **acidic conditions** (pH 0) is greater than the slope for basic conditions (pH 14). Therefore, performing the experiment under acidic conditions will make the reduction of ClO_4^- more favorable during the reaction.

(Choice B) There is no experimental data in the passage to support the idea that performing the reaction under neutral conditions (pH 7) would make the reduction of ClO_4^- more favorable. Data at pH 7 would need to be collected for comparison before any conclusion about neutral conditions could be made.

(Choice C) On the Frost diagram, the slope of the line segment between ClO_4^- and ClO_3^- at pH 14 is less steep than the slope of the line segment between ClO_4^- and ClO_3^- at pH 0. This indicates that the reduction of ClO_4^- is *less favorable* under basic conditions.

(Choice D) On the Frost diagram, the plot of the species in solution at pH 0 displays a distinctly different profile from the plot of the same species placed in solution at pH 14. Therefore, the pH of the reaction mixture *does* affect the results.

Educational objective:

Reactions associated with a larger positive standard reduction potential are more thermodynamically favorable. The standard reduction potential E° for an oxidation-reduction couple can be evaluated graphically from electrochemical data.

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